

Blood Group

Discovery - Karl Landsteiner

groups - A, B, AB, O

Universal donor - O (-ve)

Universal acceptor - AB (+ve)

No Antigen

by YM Bhende in 1952, Bombay

Lacks H antigen on RBCs

cannot take blood from anyone

protein on the surface of RBCs

Protein present - Rh(+ve)

Protein absent - Rh(-ve)

Bombay blood

Rh factor

Aorta

form L. Ventricle to Body

Largest artery & thickest

carries Oxygenated blood (from Lungs to L. Atrium)

Mitral valve - form L. Atrium to L. Ventricle

Pulmonary Vein

Tricuspid valve - form R. Atrium to R. Ventricle

Carries deoxygenated blood (from R. Ventricle to Lungs)

Pulmonary Artery

Superior & Inferior Vena Cava

From Body to R. Atrium

carry deoxygenated blood

largest vein

Heart

Pacemaker

battery-powered device that prevents the heart from beating too slowly which is placed inside the heart through surgery

Sinoatrial Node - naturally in heart

Sphygmomanometer

To measure arterial blood pressure

Arteries

carries oxygenated blood

H → Body part

Narrow, thicker

Veins

carries deoxygenated blood

Pericardium

outer membrane of heart

Coronary Artery

main blood vessel that supply oxygen rich blood to heart muscles

Systole

heart contracts → Ventricular contraction

120 mm of Hg pressure

Diastole

heart relaxes → ventricular relaxation

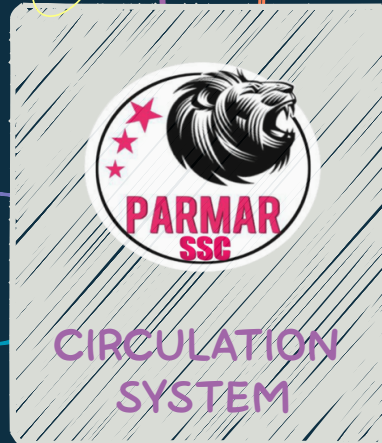
80 mm of Hg pressure

Pulse Rate

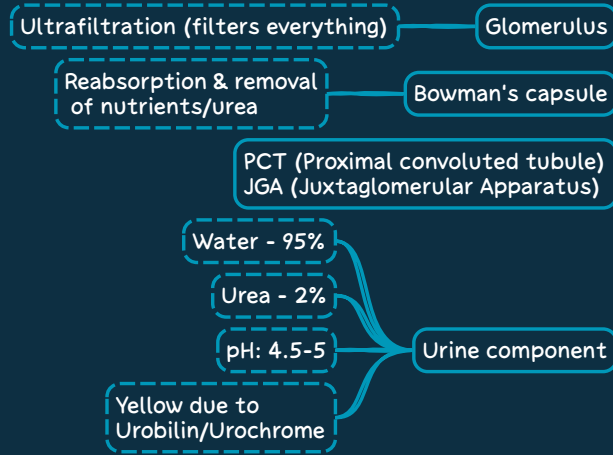
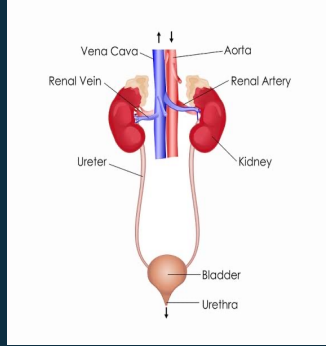
number of times, your heart beats per minute

Normal range: 60-100 beats per minute

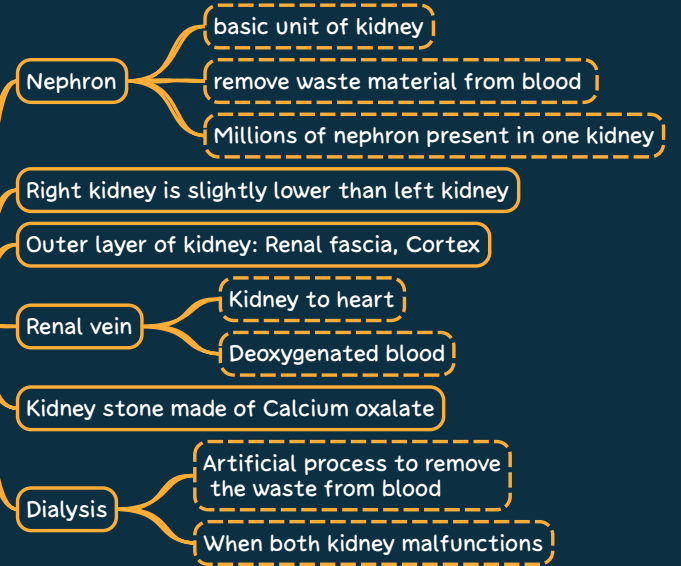
Double Circulation - passage of same blood twice through heart in order to complete one cycle



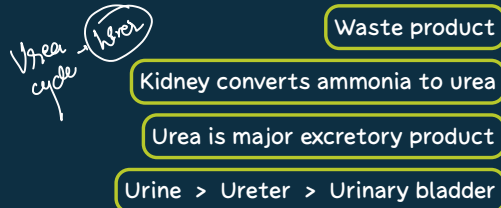
William Harvey
1578-1656



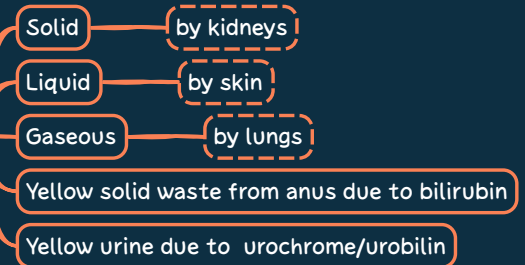
kidney



Ammonia



Waste removal



Urine

- 95% water
- 2.5% urea
- 2.5% other stuff

